

① Without pool — page-lock EVERY transfer

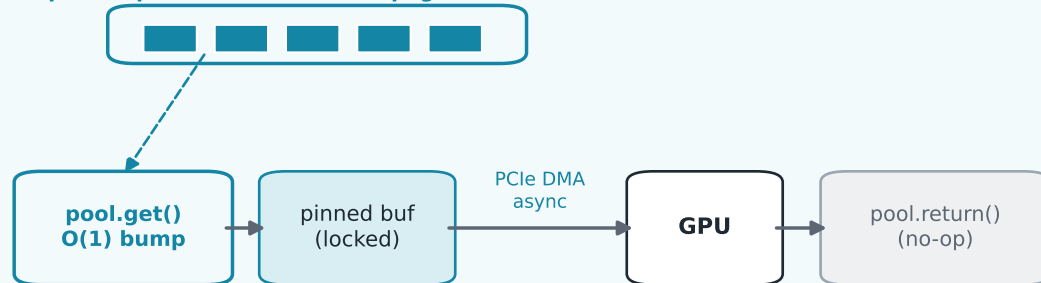
↻ repeated × N transfers



✗ every h2d/d2h pays a page-lock syscall (pin + unpin)
+ pageable mem is internally staged → can't async / full BW

② Pinned pool (ours) — lock ONCE, reuse

pinned pool (init: allocate & page-lock ONCE)



✓ zero page-lock syscalls in the transfer loop
+ pinned → DMA direct over PCIe → full BW, async

Result — wall-clock

58.7 → 9.2 ms (bcsstk30)

• 9.4 → 1.9 ms (bcsstk17)

• 1.84 → 0.91 ms (bcsstk11)